2406ET084

Roll No. _____

Course: B. Tech. Computer Engineering/
B. Tech. Computer Science and Engineering (AI/ML)
Subject: Data Structures
Subject Code: ETCS106
Semester: II

Time: 03 Hours **Max Marks: 70**

Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 02 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

Q. No.	SECTION –A (SHORT ANSWER TYPE QUESTIONS)	Marks
1. a.	Briefly describe the notions of (a) the complexity of an algorithm and (b) the space time trade-off of algorithms.	(2)
b.	What is the sparse matrices and discuss how these are represented in the memory?	(2)
c.	Define Stack ADT (Abstract Data Type). Mention the features of Stack ADT	(2)
d.	Define DEQUES	(2)
e.	The doubly linked list can be represented as follows: Left | data | right Write down the structure definition in C for the given Node of the doubly linked list.	(2)
f.	Translate following infix expression into its equivalent prefix expression: (A – B) * (D / E)	(2)
g.	Differentiate between Full Binary Tree and complete Binary tree	(2)
h.	Discuss the specific advantage of height balanced tree?	(2)
i.	Discuss representation of graph using Adjacency Matrix	(2)
j.	What do you mean by collision resolution in hashing?	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2. Define complexity of an algorithm, discuss the various asymptotic notations used to represent complexity Also analyse the complexity of the Binary Search. (10)

3. Consider the following arithmetic infix expression Q: (10)
Q: ((A + B) * C – (D - E)) ↑ (F + G)
Simulate the algorithm to transforms the infix expression Q into its equivalent Postfix expression (P) using STACK (Show the values of symbol scanned, STACK and Expression P at each and every step.)

4. A double-ended queue (deque) is a linear list where additions and deletions can be performed at either end. Represent a deque using an array to store the elements of the list and write the "C" functions for additions and deletions. (10)

5. Explain the concept of doubly linked list and also explain the operation performed on the doubly linked list along with the complexity of each operation of doubly linked list. (10)

6. Write the functions in C to perform the following operations:
   a) Deletion from the beginning in a linked list (5)
   b) Insertion at the end of the linked list (5)

7. a) Suppose the following sequences list the nodes of a binary tree T in preorder and inorder, respectively: (5)

   Preorder: G, B, Q, A, C, K, F, P, D, E, R, H
   Inorder: Q, B, K, C, F, A, G, P, E, D, H, R

   Draw the diagram of the tree

   b) Construct an expression tree for the expression (a + b * c) + ((d * e + f) * g). Give the outputs when you apply Inorder, Preorder and Postorder traversals. (5)

8. a) Explain the various representation of graph with example in detail? (5)
   b) Explain Depth First Search algorithm with example? (5)

9. Explain QUICK Sort with an example. Also write the recursive routine for the QUICK sort. (10)

===END OF PAPER===